Ethical Analysis - IMU Video

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Statement

In this report, LLMs were used solely for translation, stylistic refinement, and grammar and spelling checks. All ideas and written content originate from the authors.

Video Ethics

Using video recording in rehabilitation, education, and research presents significant benefits but also raises important ethical dilemmas related to patient privacy, informed consent, and data management. Developers of technologies involving video recording, such as in this project, must responsibly assess when and how recording is appropriate to protect participants.

In this project, video imaging is essential for accurate movement analysis. Therefore, it is critical to ensure fairness and prevent any injustice or misrepresentation in the use and distribution of videos. Clear and transparent communication with potential participants is vital. This includes explaining the study's purpose, potential benefits and risks, expected recording duration, and how videos will be collected, stored, and used.

Respecting patient autonomy is essential. All video recordings must be conducted only after obtaining informed consent through a comprehensive process that clearly outlines participants' rights, including the right to withdraw at any time without penalty. Consent should be considered ongoing, and we are responsible for maintaining its validity throughout the study.

Additionally, strict guidelines will be developed for video handling, data anonymization where possible, and protecting participant privacy to ensure ethical compliance and foster trust.

Dataethics

An important ethical consideration in this project is how the collected data is stored, collected, and managed. Even if the data itself is not highly personal, it still needs to be handled responsibly to prevent security risks or misuse. It's essential to have clear guidelines for how long the data is kept, who has access to it, and when it should be deleted. Data should only be stored for as long as it is needed, and participants should be informed about these practices. Being transparent about data handling helps promote trust and accountability within the project.

Identifying Body Measurements and Inferring Results to Others

Video-based motion tracking and IMU data can be used to extract precise body measurements, such as height, limb length, and body proportions. However, this data can reveal personal characteristics such as age, gender, body type, and skin colour. This form of implicit identification raises ethical concerns regarding privacy, discrimination, and bias. When results are generalised across individuals or groups without accounting for physiological diversity, misleading or inequitable conclusions may be reached. Therefore, body-related data must be anonymised, used only within its intended research context, and interpreted with awareness of demographic and biological variability.

Exploratory tool

As the project aims to develop an exploratory tool rather than a diagnostic system, an important ethical consideration is how the results it produces might be interpreted and relied upon. Users could potentially place undue confidence in the findings, potentially leading to misinterpretation or misuse of results. During development, we should consider how to incorporate appropriate disclaimers and contextual information to help ensure that the tool's outputs are understood as exploratory insights rather than definitive conclusions.

Liability

In the investigation of whether accurate biomechanical data can be obtained from low-cost equipment and non-complex methods, ethical reflection on liability, validation, and intention is essential. As this project evaluates system performance using non-verified motion-tracking data, the (inaccurate) results cannot be used to influence training, health, or clinical decisions. If this criterion is violated, the bioethical principles of *beneficence* (promoting the well-being of others), *non-maleficence* (avoiding harm), and *autonomy* (respecting informed decisions) are violated. Consequently, accountability must be addressed to the responsible parties (decision-makers, developers, researchers, etc.). Although the approach is not inherently unethical or dangerous, transparency about the system's limitations and intended use is vital for ensuring responsible and ethical application.